F175-SR-701
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM

F175-SW-701
SOUTHLAKES BOOSTER STATION
LV MOTOR CONTROL CENTRE

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	
Signed: C Laird	Date: 10/11/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

F175-GE-701
TEMPORARY GENERATOR
415V
200kVA
G 3~
400A
I >
1

415V FROM SMSB (BY OTHERS)
2

F175GE701-DPS
GENERATOR DECONTACTOR PANEL

SWR EARTH BAR

MCC EARTH BAR

NEUTRAL BAR

700/1A
2.5VA
(PART OF ELR PACKAGE)
NOTE 1

AFD
(NOTE 11)

400/ 1A
5VA
CL1
3

TEST

PM

PFR

2A
63A

20A
63A

MANUAL MCCB CHANGEOVER SWITCH
3 POLE
400A

400A

TEST

400/ 1A
2.5VA
5P20
3

400/ 1A
5VA
CL1.0

SD
NOTE 5

63A
100A

01F-07

415V 3NE~ 50Hz 400A 10kA for 1sec

01R-21
160A
250A
2A
20A
(NOTE 9)

01R-05
32A
160A
2A
20A
(NOTE 9)

01R-09
160A
250A
2A
20A
(NOTE 10)

01R-13
160A
250A
2A
20A
(NOTE 10)

01R-17
100A
160A
2A
20A
(NOTE 9)

F175-DB-701
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM
AND AREA L&SP DB
2000-F175-SCM-EL-40013
2000-F175-TLD-EL-40001

10kW

F175-DB-702
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM
240VAC UPS DB
2000-F175-TLD-EL-40002

ISL

F175UPS701-ISL
SOUTHLAKES BOOSTER STATION
240VAC UPS
MAINTENANCE
BYPASS SWITCH
2000-F175-SCM-EL-40014

UPS

10kW

F175-UPS-701
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM
240VAC UPS
2000-F175-TLD-EL-40002

AHF

E-SHUT
F175SR701-LCS
NOTE 8

M
3~
75kW
F175-PW-701
SOUTHLAKES
BOOSTER PUMP 1
2000-F175-SCM-EL-40011

M
3~
75kW
F175-PW-702
SOUTHLAKES
BOOSTER PUMP 2
2000-F175-SCM-EL-40012

60A
F175-HF-701
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM
ACTIVE HARMONIC FILTER
2000-F175-SCM-EL-40015

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC READING "M.E.N. LINK LOCATED IN F175-SW-701 INCOMER TIER"
   CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE". MEN LINK SHALL NOT BE INSTALLED AT THE GENERATOR.
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
4. ALL DRIVE I/O VIA COMMS NETWORK TO RTU. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F175-BLK-IC-40001.
5. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
6. MOULDED CASE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT UPSTREAM MCCB AND INCOMER REQUIREMENTS.
7. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
8. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/C WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.
9. COMPATIBLE 'TD' TOROID TO MATCH NHP ELR.
10. COMPATIBLE 'EL500S' TOROID TO MATCH AMPCONTROL ELV.
11. INCOMER ARC FAULT PROTECTION ZONE WILL TRIP UPSTREAM GENERATOR AND TRANSFORMER INCOMER ON LIGHT DETECTION ONLY. OUTGOING FEEDERS WILL TRIP TRANSFORMER INCOMER AND UPSTREAM GENSET ON CURRENT SUPERVISED LIGHT DETECTION.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 3, 4
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 3, 4
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 3, 4
- THERMAL OVERLOAD UNIT NOTE 7
- VARIABLE SPEED DRIVE NOTE 3, 4
- I > — EARTH LEAKAGE RELAY NOTE 3, 4
- FIELD-MOUNTED DECONTACTOR
- t° — MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- PFR — PHASE FAILURE RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN

HOLD

1. kVA RATING AND INTERNAL CB RATING OF GENERATOR.
2. NSR TO CONFIRM DETAILS OF NEW 415V SMSB BY OTHERS.

APPROVED FOR CONSTRUCTION

REFERENCES	DWG No	REV	DATE	REVISION	DRN	CHK	TECH APP	APP
F175-SW-701 – LV MOTOR CONTROL CENTRE INCOMER SCHEMATIC DIAGRAM SHEET 3	2000-F175-SCM-EL-40005							
F175-SW-701 – LV MOTOR CONTROL CENTRE INCOMER SCHEMATIC DIAGRAM SHEET 1	2000-F175-SCM-EL-40001							
F175-SW-701 – LV MOTOR CONTROL CENTRE ARC FLASH CONNECTION DIAGRAM	2000-F175-SCM-EL-40003							
F175-SR-701 – SOUTHLAKES BOOSTER STATION LV SWITCHROOM GENERAL ARRANGEMENT	2000-F175-DRG-EL-40001	0	20.OCT.25	APPROVED FOR CONSTRUCTION	Initial			DS

NORTHERN STAR RESOURCES LIMITED

DRA PACIFIC (PTY) LIMITED

DRAWN	[redacted]	13.MAR.2025
CHECKED	CHARLESWORTH	14.APR.2025
DESIGNED	[redacted]	14.APR.2025
TECH APP	CHARLESWORTH	20.OCT.2025
APP	HAMDORF	20.OCT.2025
CLIENT APP		

PROJECT: KCGM
FIM TSF PROJECT

TITLE: SOUTHLAKES BOOSTER STATION
F175-SW-701 LV MOTOR CONTROL CENTRE
SINGLE LINE DIAGRAM

A1 | SCALE NTS | DRG No. 2000-F175-SLD-EL-40001 | REV 0